

Manufacturing & Video-Analytics Intelligent Report

“Generate a comprehensive Manufacturing Production Performance Report using MES, Video Analytics, and InfluxDB data. Include Work Order status with planned vs. completed quantities, machine-wise utilization from CapacityAnalysis, shift-wise production from WeighingMachine, material-wise scrap from Scrap, rejection reasons from Rejected, and live machine health from InfluxDB ECE2 electrical_params (voltage, current, temperature, vibration, and cumulative cycles) and mps measurements (total count, good count, reject count, cycle count, run status, and ideal cycle time). Provide summaries, charts, key insights, machine performance analysis, and actionable recommendations.”

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Time filter applied

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Executive Summary



Matched keywords: capacity, count, cumulative, current, cycle, cycles, downtime, ece2, electrical_params, good, ideal, machine, material, mps, production, reject, run, shift, status, temperature, time, total, utilization, vibration, voltage, work_order, worker, workorder

Telemetry window: 2026-09-24T07:24:00Z → 2026-09-25T07:24:00Z (bucketed every 5m)

Work Order Status

The production schedule shows 1,000 units completed and 500 units still planned, totaling 1,500 units. Completed work orders represent about 66.7% of the total planned volume, indicating a solid but not yet full execution of the schedule. No other status categories appear, suggesting the order pipeline is currently limited to these two states. The completed quantity aligns with expected throughput, but the remaining planned quantity should be monitored to avoid potential bottlenecks. The data quality for this table is high, with only two aggregated rows.

Data preview — WorkOrder (showing 2 of 2 points/rows)

Status	sum_PlannedQty	record_count
Completed	1000.0000	1
Planned	500.0000	1

Machine Utilization Overview

CapacityAnalysis aggregates indicate five machines with recorded capacity usage. Machine 6 contributed the highest summed capacity of 69 units, followed by Machine 1 with 62, Machine 3 with 41, Machine 2 with 39, and Machine 8 with 20. This distribution highlights that Machine 6 is the most heavily utilized, while Machine 8 shows relatively low activity, which may warrant a review of scheduling or maintenance. The record count per machine ranges from 1 to 3, reflecting varying observation windows. Overall utilization appears uneven across the fleet.

Data preview — CapacityAnalysis (showing 5 of 5 points/rows)

MachineId	sum_CapacityId	record_count
6	69	3
1	62	3
3	41	2
2	39	2
8	20	1

Shift-wise Production (WeighingMachine)

No data was returned for the WeighingMachine table, which should contain shift-level production totals. This absence suggests a data collection or integration gap; the shift performance cannot be evaluated without these records. Verify that the weighing stations are correctly publishing to the MES database and that the query window aligns with active shifts.

Material-wise Scrap Summary

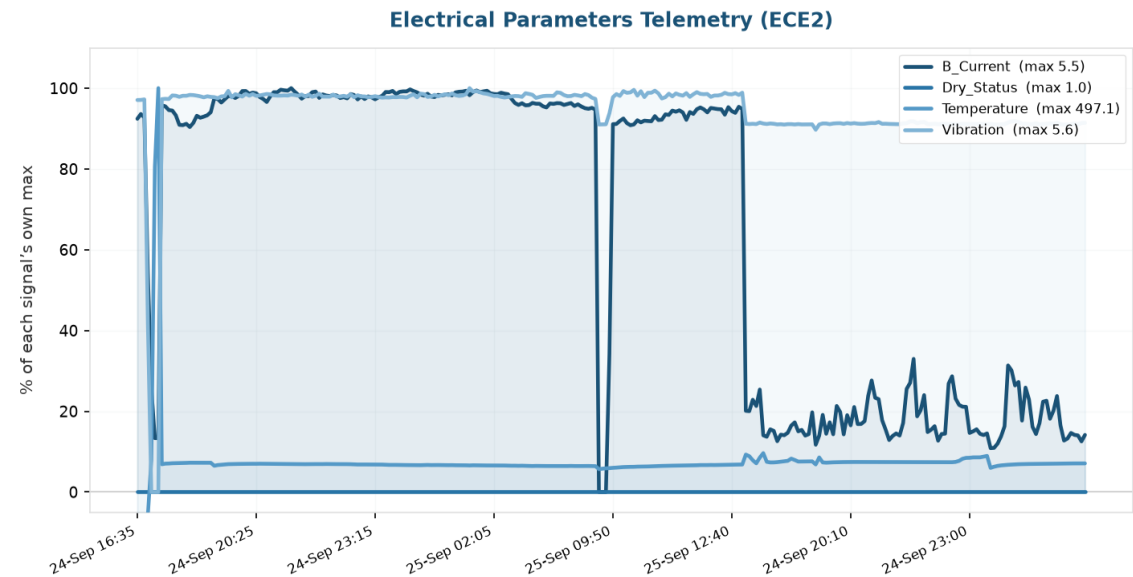
The Scrap table did not appear in the retrieved dataset, so material-specific scrap quantities are unavailable. Without this information, it is impossible to pinpoint high scrap materials or assess the effectiveness of waste reduction initiatives. Check the scrap logging process and ensure the Scrap table is populated and accessible.

Rejection Reasons Analysis

No records were returned from the Rejected table, meaning rejection cause data is missing. This prevents analysis of defect patterns or root cause identification. Confirm that rejection events are being captured in the MES system and that the query targets the correct time frame.

Electrical Parameters Telemetry (ECE2)

Telemetry for electrical_params over the last 24 hours shows 272 sampled points. Current (B_Current) ranged from 0 A to 5.515 A with an average of 3.629 A, but the latest reading dropped to 0.78 A, indicating a falling trend. Temperature exhibited extreme sensor readings (min -126.6 °C, max 497.1 °C) with an average of 35.8 °C and a rising trend, suggesting possible sensor drift or calibration issues. Vibration stayed high, averaging 5.33 units, max 5.65, and is slowly decreasing. Dry_Status remained constantly at 0, indicating no dry-run condition. Voltage and cumulative cycles were not present in this measurement, pointing to a configuration gap for those metrics.



Telemetry preview — electrical_params (showing 10 of 272 points/rows)

_time	device_id	B_Current	Dry_Status	Temperature	Vibration
2026-09-24 11:05:00+00:00	00WH9OUI	5.097014925373132	0.0	-123.12985074626867	5.482720859015166
2026-09-24 11:10:00+00:00	00WH9OUI	5.162000000000004	0.0	-120.10533333333334	5.483856337865194
2026-09-24 11:15:00+00:00	00WH9OUI	5.115333333333331	0.0	-126.57466666666677	5.491382967631022
2026-09-24 11:20:00+00:00	00WH9OUI	2.9991452991453014	0.0	-28.394871794871847	2.351515965583997
2026-09-24 11:25:00+00:00	00WH9OUI	1.4186666666666692	0.0	39.261333333333337	0.0
2026-09-24 11:30:00+00:00	00WH9OUI	0.7366666666666682	0.0	400.29733333333333	0.0
2026-09-24 11:35:00+00:00	00WH9OUI	0.7358974358974367	0.0	497.1324786324784	0.0
2026-09-24 11:40:00+00:00	00WH9OUI	5.281609195402305	0.0	34.235632183908066	5.49489330971378
2026-09-24 11:45:00+00:00	00WH9OUI	5.266666666666664	0.0	34.955333333333336	5.49951594988505
2026-09-24 11:50:00+00:00	00WH9OUI	5.216000000000011	0.0	35.123999999999997	5.496886746088664

MPS Production Telemetry

No telemetry data was found for any MPS measurement (total count, good count, reject count, cycle count, run status, ideal cycle time) across all machines (BM-01 to BM-11). This complete lack of data suggests that the MPS data stream is not being ingested into InfluxDB, or the measurement names are mismatched. Without these metrics, real-time production efficiency and cycle performance cannot be assessed. Investigate the data pipeline and ensure the MPS tags/fields are correctly written.

Appendix — Data Quality

Table	Records	Completeness	Outliers	Score
attendances	1	100.0%	0	Good
employees	7	100.0%	0	Good
CapacityAnalysis	5	100.0%	0	Good
MachineMaster	25	89.3%	0	Fair
WorkOrder	2	100.0%	0	Good
WorkOrderLog	1	75.0%	0	Needs review
MachineMaterialMapping	8	100.0%	0	Good
Machine	10	90.0%	0	Fair
MachineFGMapping	39	68.6%	0	Needs review

LLM Usage & Cost

Model	Input tokens	Output tokens	Total tokens	Estimated cost
groq/openai/gpt-oss-120b	4,769	2,040	6,809	N/A

Aggregation Fallbacks

[mes.Machine] Aggregate query failed (('42000', '[42000] [Microsoft][ODBC Driver 18 for SQL Server][SQL Server]Operand data type nvarchar is invalid for sum operator. (8117) (SQLExecDirectW)')) - served raw rows instead.

InfluxDB Measurements With No Data

[influxdb.BM-01] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-02] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-03] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-04] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-05] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-06] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-07] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-08] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-09] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-10] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-11] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows